

Baekseok Kim

+1-725-275-1910 | kimb52@unlv.nevada.edu | Research Demos

Skills

RL & Control: End-to-end RL, residual/hybrid policy learning, reward and observation design, sim-to-real, whole-body control, MPC, loco-manipulation

Programming: Python, PyTorch, NumPy, C/C++ , MATLAB

Simulation & Physics: MuJoCo, Isaac Lab, legged robot dynamics modeling, MATLAB/Python simulators

Hardware: Humanoid and bipedal legged platforms, force-augmented control, sensor integration, on-robot policy validation

Experience

PhD Research Assistant, University of Nevada, Las Vegas – Las Vegas, NV Mar 2022 – Present

- Developed end-to-end RL pipelines for bipedal locomotion on a humanoid leg platform, including policy training in MuJoCo/Isaac Lab, reward design, and sim-to-real deployment on hardware (demo)
- Designed hierarchical control architecture with pre-IK residual RL for humanoid loco-manipulation, integrating model-based whole-body control with learned residuals on custom bipedal hardware (demo)
- Built hybrid residual RL and model-based control stacks on the Stilt-bot underactuated bipedal platform for stable walking policies validated on physical hardware (demo)
- Delivered whole-body humanoid motion control as lead control engineer on Team Avatar-Hubo for the ANA Avatar XPRIZE Finals, integrating locomotion and upper-body coordination on real hardware (demo)
- Evaluated learned and model-based controllers in simulation and on hardware, iterating on observation design, dynamics modeling, and disturbance recovery behavior

Robotics Engineer / Project Manager, Global Zeus (ZEUS Co., Ltd.) – Republic of Korea Dec 2018 – Feb 2022

- Developed real-time robot dynamics, kinematics, and motion-control software with Python/OpenGL simulation tools for controller debugging and performance evaluation (demo)

Research Assistant, Hanyang University – Seoul, Republic of Korea Mar 2016 – Feb 2019

- Built MATLAB dynamics simulation and stability-analysis tools for a skiing humanoid robot and validated controllers on real-world slopes (IROS 2018)

Selected Publications

- **B. Kim, P. Oh**, “A Hierarchical Control Architecture for Humanoid Loco-Manipulation with Pre-IK Residual RL,” *52nd ICSEng*, Las Vegas, NV, USA, 2026 (Accepted)
- **Y. Noh, B. Kim, P. Oh**, “A Lightweight AI-Ready Humanoid Leg Platform for Hybrid Control Using Residual RL,” *2026 IEEE CCWC*, 2026
- **B. Kim, P. Oh**, “A Simplified Underactuated Platform for AI-Ready Bipedal Walking Control: The Stilt-bot,” *2025 ACC*, 2025
- **B. Kim, P. Oh**, “Force-augmented LIPM-MPC for Stable Humanoid Loco-Manipulation,” *2026 ACC*, 2026
- **B. Kim et al.**, “Approach of Team Avatar-Hubo to the ANA Avatar XPRIZE Finals,” *2024 IEEE RO-MAN*, 2024
- **J. Han et al.**, “Research on Carved Turns of a Skiing Humanoid Robot on a Real-World Slope,” *2018 IEEE/RSJ IROS*, 2018

Selected Honors

- Humanoid Robot Autonomous Winner, *2023 IEEE-RAS International Conference on Humanoid Robots Competition*
- UKC 2023 Young Scholar Award, *US-Korea Conference on Science, Technology, and Entrepreneurship*

Education

University of Nevada, Las Vegas – PhD in Mechanical Engineering (Advisor: Paul Y. Oh; GPA: 4.0/4.0) Dec 2026
Hanyang University – MS in Interdisciplinary Robot Engineering Systems (GPA: 4.0/4.0) Feb 2019
Sungkyunkwan University – BS in Mechanical Engineering (Major GPA: 3.43/4.0) Feb 2017